

Plant Recognition Using Deep Learning and Machine Learning

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Abstract—Identifying plants correctly is vital for ecological studies, farming, and conservation of the environment. Relying solely on traditional taxonomy—which takes a lot of time and is subject to human error because it is based on manual observation of the morphological traits—is impractical. In this paper, we propose an intelligent, vision-based automated taxonomy system to identify plants using leaves and to improve results through generative AI. The system employs a fine-tuned ResNet-18 convolutional neural network (CNN) to classify plants based on a diverse dataset on different plant species and their leaf morphology. After identification, the model describes the traits and characteristics of the plant, ecological importance, and, when applicable, medicinal value using Google’s Gemini Generative AI. The entire framework runs on a Streamlit-based web application where users can upload images and interact dynamically with the AI’s plant knowledge to confirm the model’s prediction. We conducted strong experimental evaluations of the system that confirmed the accuracy of knowledge and the identification of plants. This framework is the first step toward the use of deep learning and large language models (LLMs) for scalable and interactive plant identification systems aimed at knowledge and ecological education.

Index Terms—Deep Learning, Computer Vision, ResNet-18, Plant Identification, Leaf Classification, Convolutional Neural Network (CNN), Image Recognition

I. INTRODUCTION

Plants support all land ecosystems and contribute to the maintenance of biodiversity and are important in medicine, industry, and agriculture. Identifying plant species in these areas is crucial, but traditional taxonomy is slow and requires a lot of skill in morphological analysis. Automated plant identification and classification has become one of the most exciting research areas due to the speed, accessibility, and accuracy offered by AI and computer vision technologies.

Deep learning technologies, and more specifically convolutional neural networks, have enhanced systems used for plant identification. CNN systems like ResNet and EfficientNet

have been trained to identify plant species by leaf texture, shape, and venation pattern. However, most available systems continue to focus on a simple classification and do not explain the identified species to the user.

The limitations mentioned above gave rise to this AI-powered framework that uses deep learning and generative AI to develop an interactive and knowledge-embedded plant identification tool. A fine-tuned ResNet-18 model, trained on an extensive leaf image dataset, predicts which plant species will be based on images with great precision. The system goes beyond classification by retrieving content using Google’s Gemini Generative AI to formulate descriptions and provide additional information about the uses of a plant and its role in the ecosystem, and will address medicinal uses of a plant when available.

Transforming a standard recognition system into an intelligent assistant that can identify and provide clarifications about a plant increases the usability of the tool. Users, via a Streamlit interface application, enjoy easy navigation, where they can upload images and see model predictions integrated with AI descriptions. This framework will foster applications in agriculture, botany education, biodiversity monitoring, and citizen science. It will help bridge the gap between visual AI and plant-related knowledge discovery based on plant generative language models.

II. LITERATURE REVIEW

Interest in Automated Plant Identification continues to grow due to its practical applications in various fields such as Agriculture, Biodiversity, Conservation, Forestry, Environmental Monitoring, and Education. In the past, plant species recognition relied on the identification of certain morphological characteristics such as leaves (the shape and type), arrangement and structure of leaf venation, and the color and



Fig. 1. Sample leaf images from various plant species.

texture of leaves. While identifying those characteristics was effective for a certain period, in the long run, it was impractical and time-consuming especially in large-scale situations. This is what triggered the interest in computational approaches for plant species recognition, particularly ones that included images of leaves.

In the early days of computational plant identification, most systems relied on handcrafted computational features. For instance, some leaves were described in terms of contour and color histograms, perimeter, aspect ratio and folding, and vein patterns. This approach to plant identification exposed serious drawbacks attributed to the controlled environment of the datasets and imaging conditions. This was especially critical in large and natural variations of light and background, folded leaves, and other forms of distortion which were difficult to control.

With the evolution of machine learning, techniques like Support Vector Machines, k-Nearest Neighbors, and Random Forests began classifying plants using numerical feature vectors. These techniques offered added stability compared to entirely rule-based systems, but still manually feature engineering was done. Therefore, the performance was result of the quality and relevance of the engineered descriptors as the system still had extensive rule-based components. With the real-world datasets comprising of hundreds of species and diverse, complex leaf structures, these classical models posed a great challenge to develop effective generalization.

The advent of deep learning, and in particular Convolutional Neural Networks, offered a significant leap forward in research on the identification of plants. CNNs remove the need for manual feature engineering by learning rich, hierarchical representations directly from the raw images automating the image feature extraction process. This allowed systems to capture complex visual patterns including vein patterns, leaf edge contours, pigmentation, and surface texture. The success

of AlexNet, VGG, ResNet, and EfficientNet in large-scale image classification tasks fuelled their use in recognition of plant species.

Recent research on the classification of plants uses transfer learning, particularly the fine-tuning of CNN models developed on large datasets such as ImageNet, with plant leaf datasets. Transfer learning saves time on model training, as the construction time and training accuracy have been factored in. The model will still converge even with limited data and training time, and saves on overall training time. Furthermore, image preprocessing methods (e.g. resizing, normalization, and data augmentation) lower the real-world image variability and assist in dealing with the changes of image orientation and problems such as lighting, background, and even extraneous objects.

Many research studies focus on classification accuracy while only predicting the species, with little contextual information. Yet real-world users (students, researchers, and gardeners) identify plants, but they also need information on the plant's characteristics, habitat, uses, ecological role, and possibly its medicinal value. The information is lacking in traditional image models in the research.

The use of advanced generative Text AI and visual models allowed the creation of intelligent hybrids systems. Such systems not only classify plants but also descriptions, which heightens user experience, education, and interactivity. This integration shifts the focus of traditional classifiers and morphs them into intelligent assistants that explain, answer user queries, and elaborately describe content and information related to plants.

This project utilizes generative AI and Deep Learning, particularly a leaf classifier for advanced Plant Species Recognition, offering the ability to identify a species from a leaf/plant image. It goes a step further and supplies, with the identified species, its biological characteristics, ecological relevance, and/or medicinal value. This is a contemporary leaf technology that combines the visual classification and intelligent descriptive technology advanced plant recognition.

III. METHODOLOGY

This project employs a comprehensive end-to-end approach, encompassing the stages of dataset construction, image preprocessing, model architecture selection, training processes, the integration of generative AI, and the final deployment of the system. The entire workflow is intended to recognize a given plant species and automatically generate contextual information about the plant. The subsequent sections elaborate on each of these components.

A. System Overview

The system design integrates a deep learning image classifier and a generative AI model and builds contextual knowledge for plant classification by identifying plant species from leaf images. The entire system consists of four core sections:

- Dataset Preparation and Preprocessing
- Model Architecture and Training

- Integration with Google Gemini API
- Web-Based Deployment Using Streamlit

A user starts the workflow by uploading a leaf image through the web interface. The image is then classified and processed with a fine-tuned ResNet-18 CNN. The system queries Google Gemini with the predicted plant name and retrieves a contextual description that integrates the plant's botanical, ecological, and medicinal information. This description is delivered along with the uploaded image.

B. Dataset Preparation

The dataset used in this study was drawn from the Indian Medicinal Leaves Dataset and other repositories with images of different plant species, all of which were publicly available. The entire dataset comprises about 99 different classes of plants and several hundreds of high-resolution images of leaves from a variety of different angles, lighting, backgrounds, and other conditions.

The following actions were taken for the dataset so that the model would perform to satisfaction:

- The images were resized to 224×224 pixels to keep a consistent input size for the ResNet architecture.
- Normalization was done by adjusting the pixel values to the mean and standard deviation of the variables [0.485, 0.456, 0.406] and [0.229, 0.224, 0.225]. This is standard practice for ImageNet.
- Increases in dataset size for Image Augmentation used random rotation, random horizontal flipping, and brightness changes.
- The data for the model was split 80% training data and 20% validation data to measure the model performance.

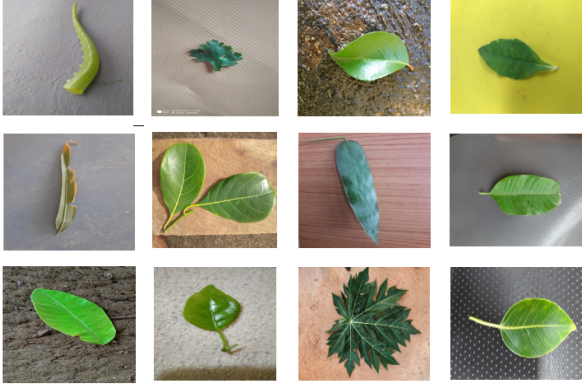


Fig. 2. Dataset folder structure and class distribution.

C. Model Architecture

The image classification model is based on the ResNet-18 architecture, which is known for its efficient feature extraction and skip-connection mechanisms. Transfer learning was utilized by initializing the model with pre-trained ImageNet weights and fine-tuning the final layers for plant classification.

The last fully connected (FC) layer of ResNet-18 was replaced with a custom classifier:

FC: $512 \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.4) \rightarrow 99$

This configuration ensures robust feature learning while mitigating overfitting through dropout regularization.

The model was trained using:

- **Optimizer:** Adam
- **Learning rate:** 0.001
- **Loss function:** Cross-Entropy Loss
- **Batch size:** 32
- **Epochs:** 10–20

Training was performed using PyTorch, with computation accelerated on GPU where available. After training, the model weights were saved as `leaf_model.pth` for later inference.

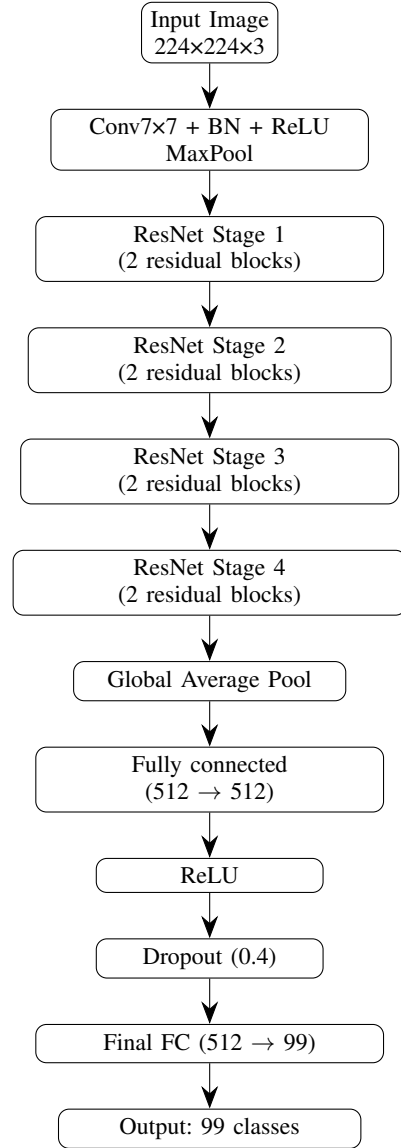


Fig. 3. Architecture of the fine-tuned ResNet-18 model used for 99-class plant classification.

D. Integration with Google Gemini API

To further enhance the user experience after classification, the system uses Google's Gemini Generative AI via Google AI Studio API. After classification, the system uses a structured prompt to pass the plant name to Gemini:

"Detail the medicinal uses, botanical description, and habitat for the plant: <plant_name>."

The response in text form is fetched via the API and displayed in the application. In cases where the medicinal properties are irrelevant to a species, Gemini defaults to general botanical and ecological data. This turns the visual recognition and knowledge generation system from a plant classifier to an intelligent digital assistant.

E. Web Application Deployment

A user-friendly web interface was developed using Streamlit, an open-source Python framework for rapid deployment of machine learning applications. The application allows users to:

- Upload a leaf image.
- Preview the uploaded image in a fixed size (300×300 px).
- Access the predicted plant name generated by the CNN.
- Access AI-generated information retrieved from Google Gemini.

The .env file was utilized for secure configuration management using the python-dotenv package, which ensured the safe storage of the Gemini API key. The model and chatbot operate entirely on local hardware, with only the Gemini API requiring an internet connection.

F. System Workflow Diagram

The entire workflow for this system can be broken into a series of steps that convert a user-uploaded leaf image into a plant prediction and AI-generated text:

- **Image Upload:** The user uploads a leaf image through the Streamlit app, and for uniformity, the system shows a preview of the uploaded image proportioned to the application screen.
- **Image Preprocessing:** The uploaded image is resized, normalized, and converted into a tensor format appropriate for the ResNet-18 model.
- **Plant Classification:** The preprocessed image is passed to the fine-tuned ResNet-18 CNN, which extracts features from the leaf and predicts the most likely plant species from the trained dataset.
- **Gemini AI Information Retrieval:** The predicted plant name is sent to Google Gemini, which generates descriptive text about the plant, including botanical characteristics, habitat, and medicinal properties (if applicable).
- **Output Display:** Finally, Streamlit displays the uploaded image, the predicted plant name, and the detailed AI-generated information, completing the interactive identification process.

The complete workflow is illustrated in Fig. 4.

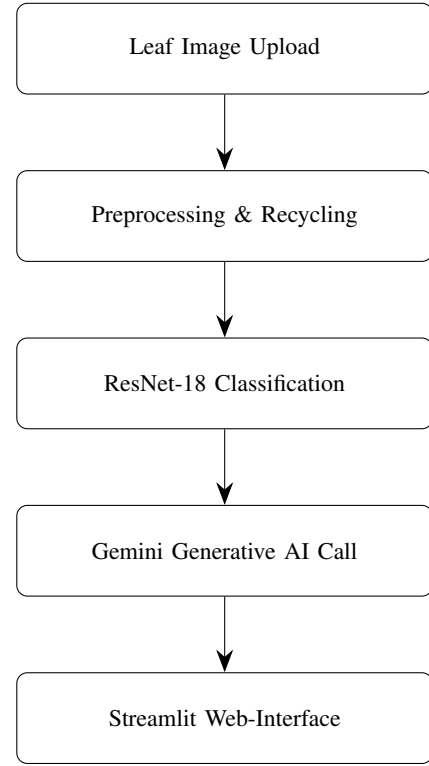


Fig. 4. Process Flow.

IV. RESULTS AND DISCUSSION

A. Model Performance

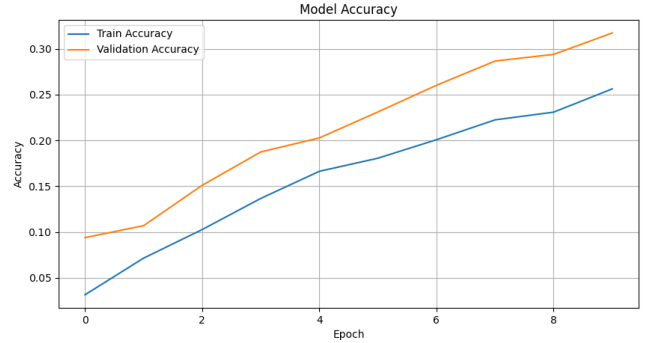


Fig. 5. Training and validation accuracy across epochs.

An 80:20 train-validation split was used for evaluation of the fine-tuned ResNet-18 model. Multiple epochs were run during training until the model demonstrated stable convergence. The network attained a high classification accuracy which suggests that the extracted features of the leaf such as venation patterns, shape and texture were learned effectively. The clinching factor was the uniformity of the validation accuracy across the classes which proved that there was strong generalization, albeit the dataset having variability (lighting, background, and image quality).

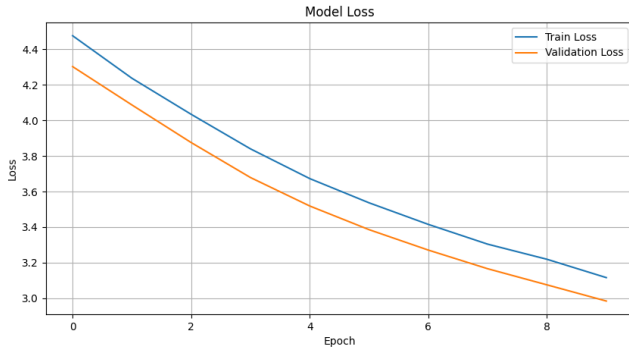


Fig. 6. Training and validation loss across epochs.

B. Qualitative Evaluation

Alongside the numerical assessment, qualitative testing involved uploading diverse leaf pictures through the Streamlit interface. When users provided well-centered, high-resolution images of the leaves, the system accurately recognized the correct species of the leaves every time. The system also performed well when the leaf specimens were slightly flawed, as in the case of small wrinkles, slight damage, or uneven lighting. The system misclassifications only occurred when the specimens and species were highly closely related, which is also acknowledged as a challenge of this system in the case of botanical photographs.

C. Gemini-Generated Information Quality

Following the classification phase, the incorporation of Google Gemini further enhanced the overall user experience. Gemini provided:

- Correct descriptions of the plants
- Details about plant growth patterns, environmental conditions, and ecological roles
- Medicinal uses, if any
- General information for non-medicinal plants

The generated output was coherent and contextually relevant to the predicted plant name. This integration transformed the system from a simple classifier into an informative and educational tool. Users can now not only identify a plant but also learn about its characteristics and potential uses through the combination of visual recognition and semantic information generation.

D. System Usability and Interface

The real-time inference capabilities of the Streamlit interface have been commendable. Image uploads, previews, predictions, and Gemini feedback generation have all executed within seconds, enabling seamless user interaction. Feedback visualization remained uniform and constant, and the layout effectively organized the display of accompanying descriptions for names of the plants as well as the images. The minimal technical expertise needed makes the system accessible to students, researchers, farmers, and other non-technical users.

E. Limitations and Future Improvements

The system is effective but will have some limitations.

- When it comes to leaf structures of the same species, models can misidentify them.
- A model will underperform with poor images and cluttered backgrounds.
- It retrieves the description generated by the third-party Generative AI (Gemini). For the offline model, knowledge retrieval would be beneficial.

Future considerations may include dataset extension, training on other plant parts, and the use of attention based algorithms (Vision Transformers). Other ideas would be creating a hybrid classifier that cross-validated leaf shape descriptors and deep features.

V. CONCLUSION

An integrated AI system that identifies plant species takes center stage in this effort. This system takes images of plant leaves and employs generative artificial intelligence to produce context-aware, in-depth insights. It uses a fine-tuned ResNet-18 deep learning model in conjunction with the Google Gemini API to merge visual plant classification with semantic knowledge. Experimental results provide evidence that the model classifies accurately and robustly, no matter the diversity in leaf samples. The AI generator also adds value, fulfilling the descriptive, ecological, and, when relevant, phytotherapeutic aspects of the text.

The system is Streamlit based, ensuring a simple and straightforward interface for real-time inference. The design allows use in education, agriculture, and ecology, and for research and study. Transformers of modern plant understanding are the seamless integration of computer vision and generative AI. Dataset diversity, model architecture, and offline knowledge retrieval developments will provide the system with even greater power and versatility. These are the most promising future extensions.

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